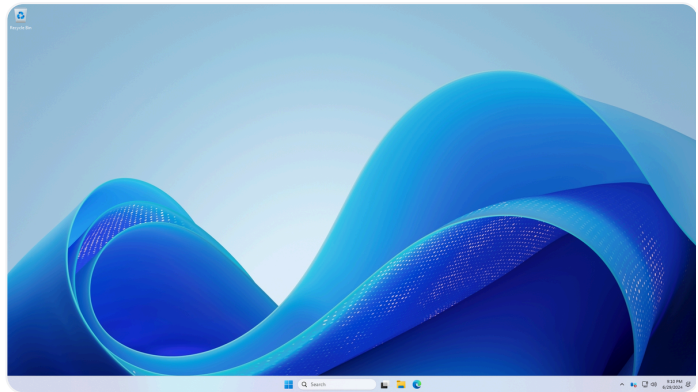
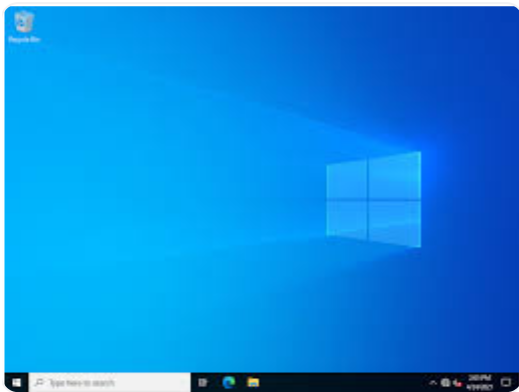


How to build a Microsoft AD/Exchange Environment with a PFSense Firewall

- First Install Windows Server Standard/Datacenter 2016/2019/2022/2025 for our Active Directory and (DHCP) Server VM or Bare-Metal System



Building Active Directory Users and Computers

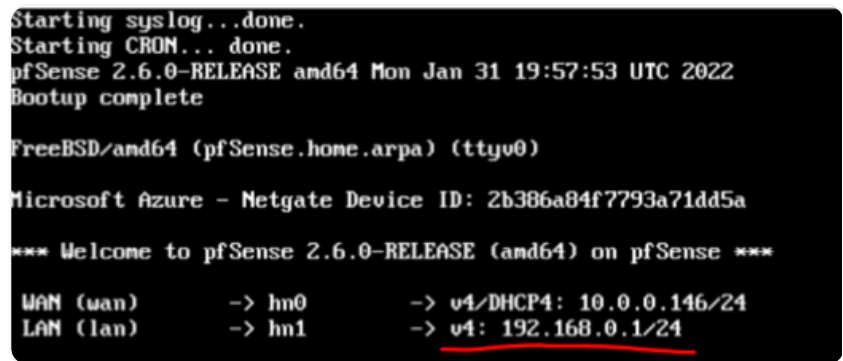
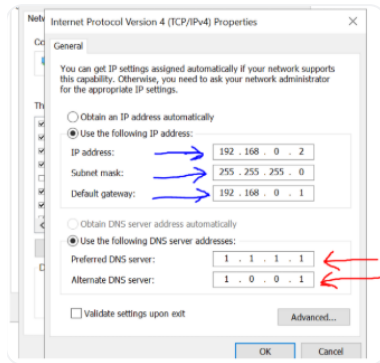
Step 1:

(LEFT IMAGE)

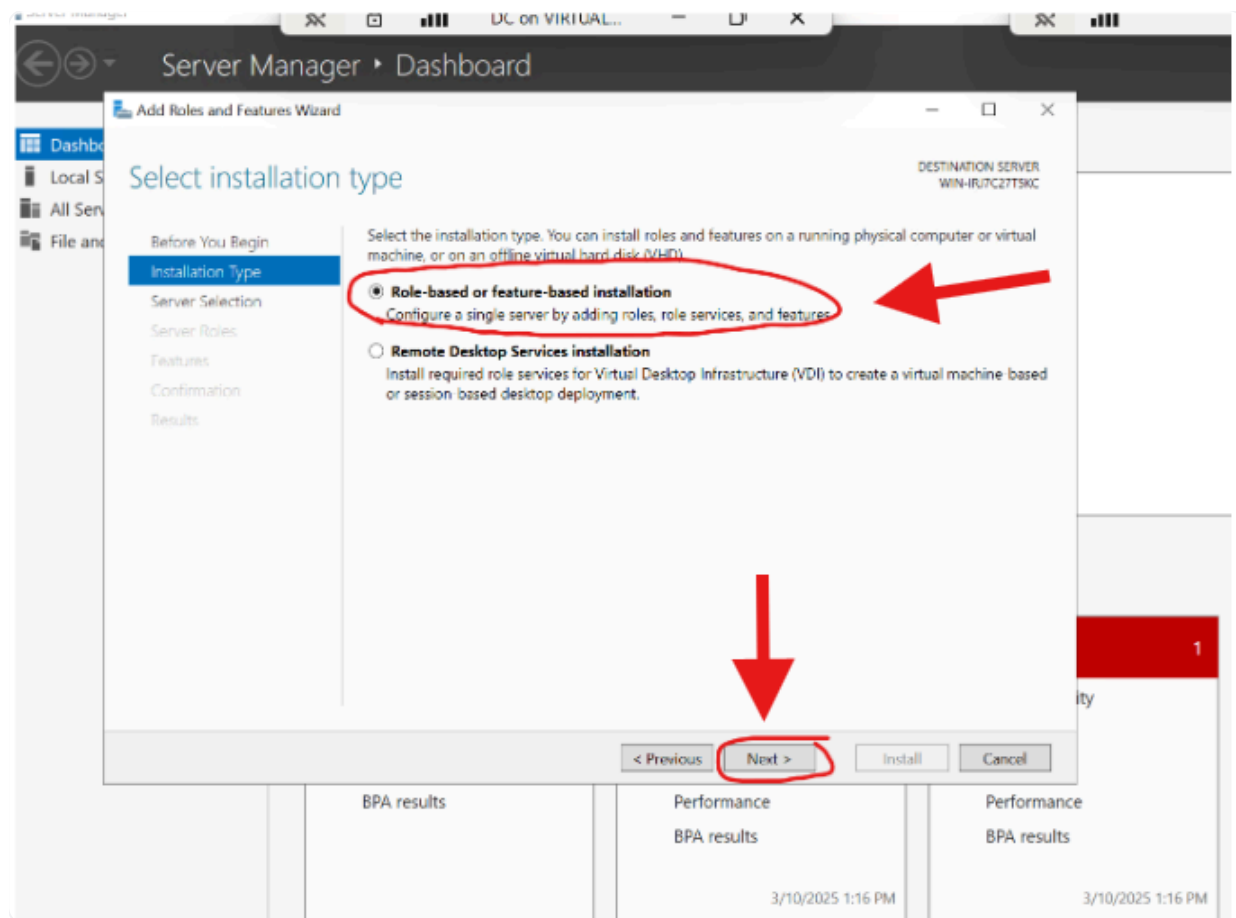
- Give your Machine a Static IP/subnet/ and Default gateway (BLUE)-(Usually default gateway of your Network Router or Network Firewall) and Live DNS from 1.0.0.1 and CloudflareDNS 1.1.1.1 (RED)

(RIGHT IMAGE)

- Here we are using a **PFsense Firewall Appliance** to Provide our AD With a Default Gateway (RED LINE) But this can be done from your Home/Business Router with the Creation of a DNS or DHCP in its browser Homepage.

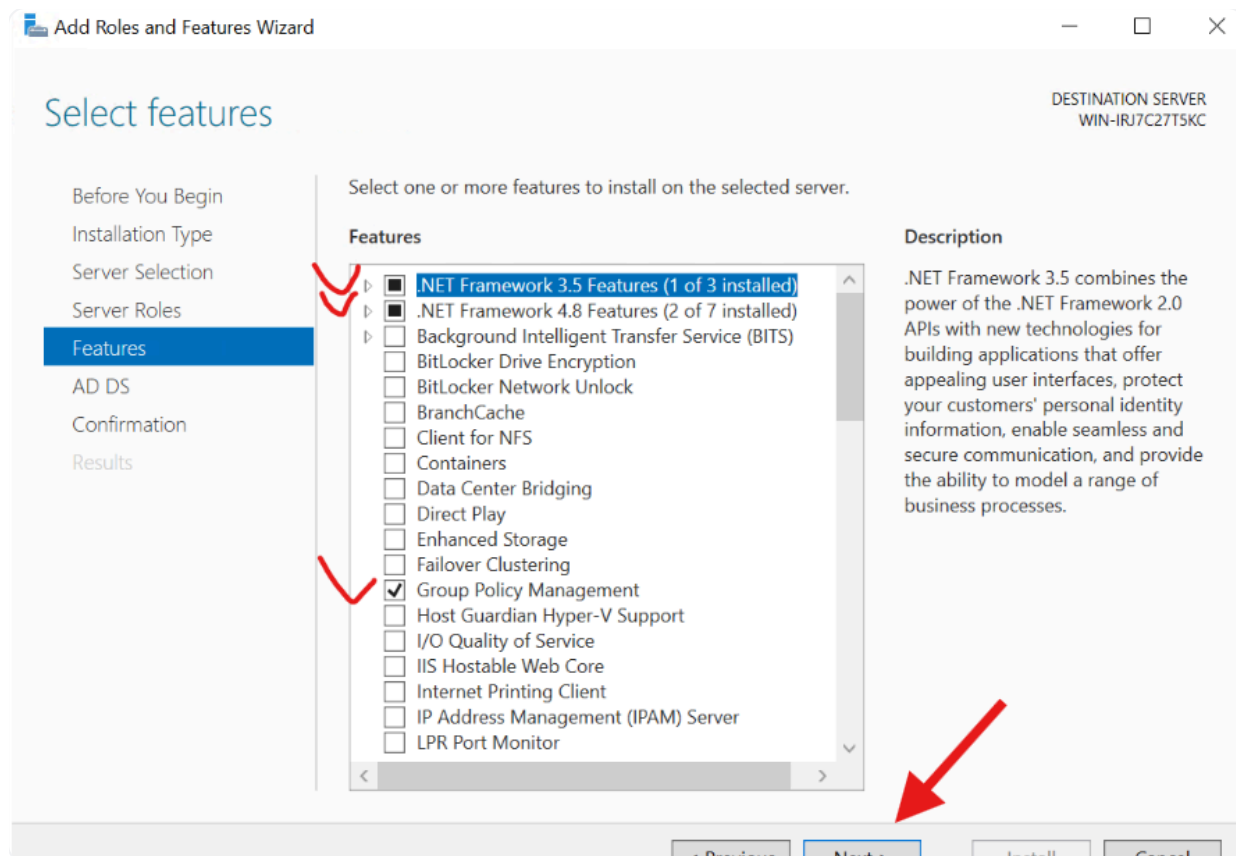


After signing in to Windows server 2022/2025 Start by opening “Server Manager” And Click on “Manage” In the Top Right Tool bar “Add Roles and Features”. Then Click Next

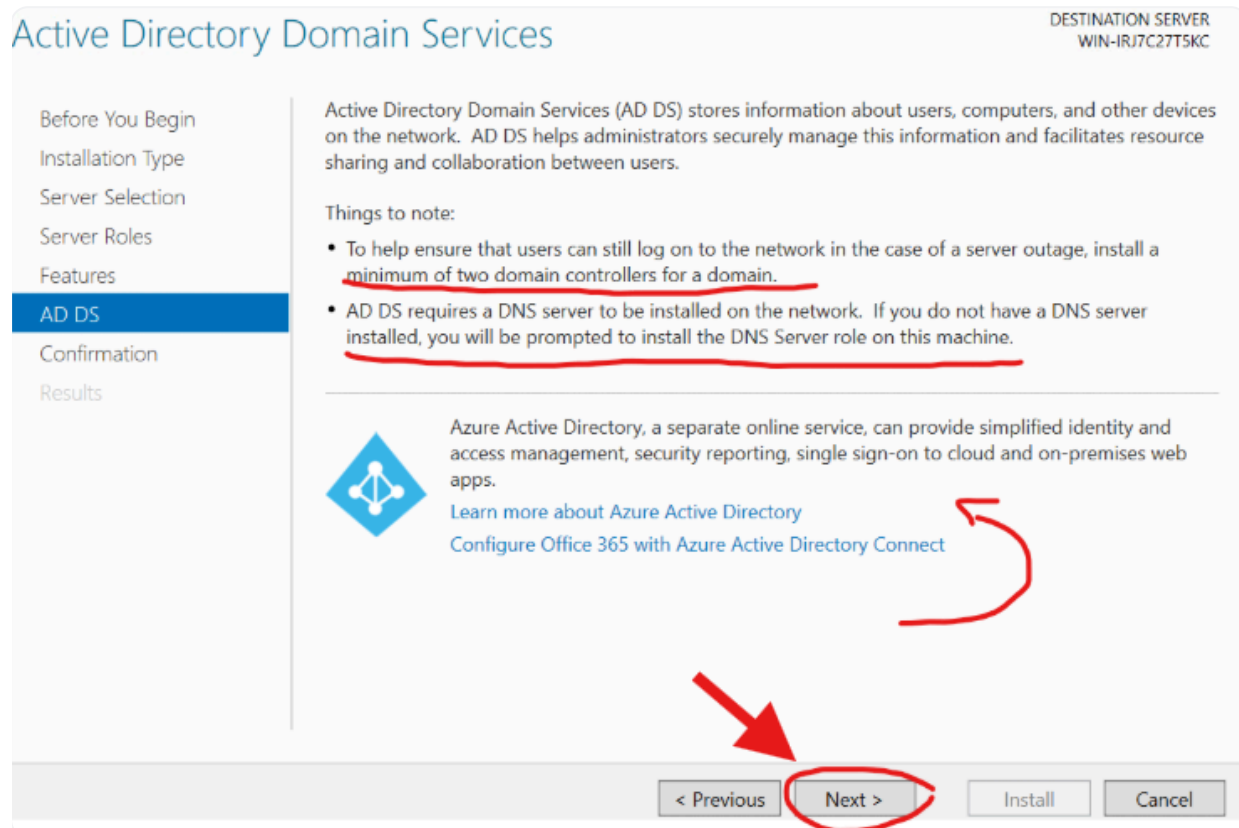


Next check that “**NET framework 3.5/4.8**” will be installed alongside “**Group Policy Management**”

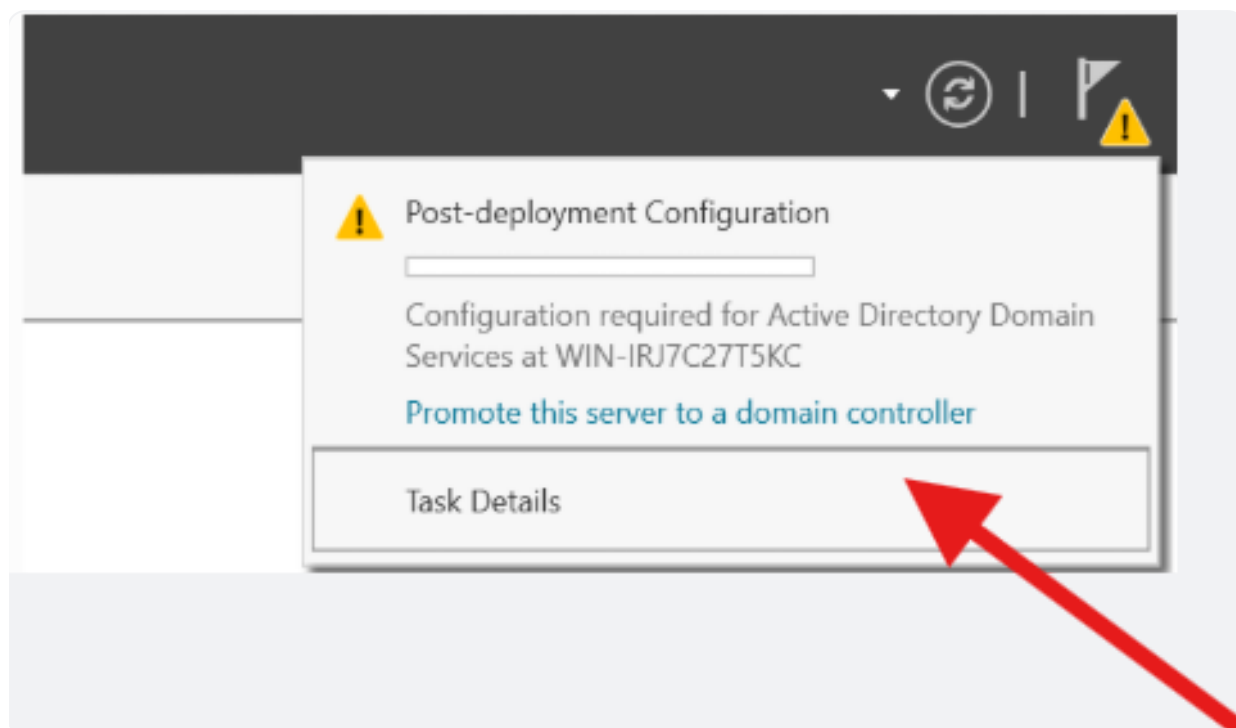
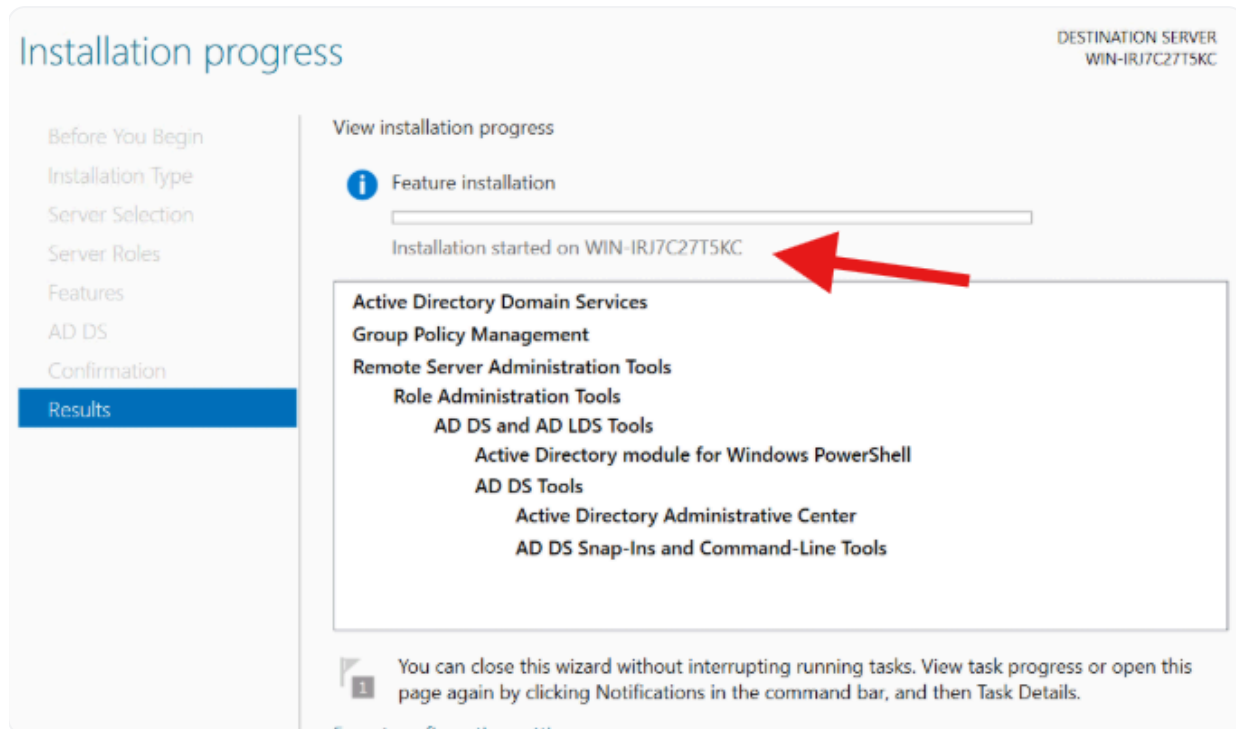
.NET Framework provides a foundational platform for Windows Server to run, manage, and secure applications, acting as a bridge between software and the OS. Providing Essential Services :Memory management, security and connectivity so new and legacy applications run reliably



- Before confirming the installation Remember to Check your DNS Server was installed so your AD has something to connect to every other machine.



- Now Install your AD and allow it to perform a Restart in the checkbox.



- Add your New Forest Domain with your Registered Domain for your company to use properly with regulation. (Forest: - is the top-level container and security boundary in Microsoft AD) sharing across all internal domains while isolating them from external entities

<https://learn.microsoft.com/en-us/troubleshoot/windows-server/active-directory/naming-conventions-for-computer-domain-site-ou>

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the window icon, the text 'Active Directory Domain Services Configuration Wizard', and standard window controls (minimize, maximize, close). The main area is titled 'Deployment Configuration'. In the top right corner, it says 'TARGET SERVER WIN-IRJ7C27T5KC'. On the left, a vertical list of steps is shown: 'Deployment Configuration' (highlighted in blue), 'Domain Controller Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is divided into two sections. The first section, 'Select the deployment operation', contains three radio button options: 'Add a domain controller to an existing domain', 'Add a new domain to an existing forest', and 'Add a new forest' (which is selected). The second section, 'Specify the domain information for this operation', contains a label 'Root domain name:' followed by a text input field. Two red arrows are drawn on the image: one points to the 'Add a new forest' radio button, and the other points to the 'Root domain name' text input field. At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'. A link 'More about deployment configurations' is located at the bottom left of the main content area.

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
WIN-IRJ7C27T5KC

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous Next > Install Cancel

- Add a solid and Secure password in case you need to use the Restore mode for failing/ Compromised Domain Controller

Domain Controller Options

TARGET SERVER
WIN-IRJ7C27T5KC

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous

Next >

Install

Cancel

DNS Delegation is only Necessary if you're adding on to another DNS environment.

DNS Options

TARGET SERVER

 A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found... [Show more](#)

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify DNS delegation options

☐ Create DNS delegation

[More about DNS delegation](#)

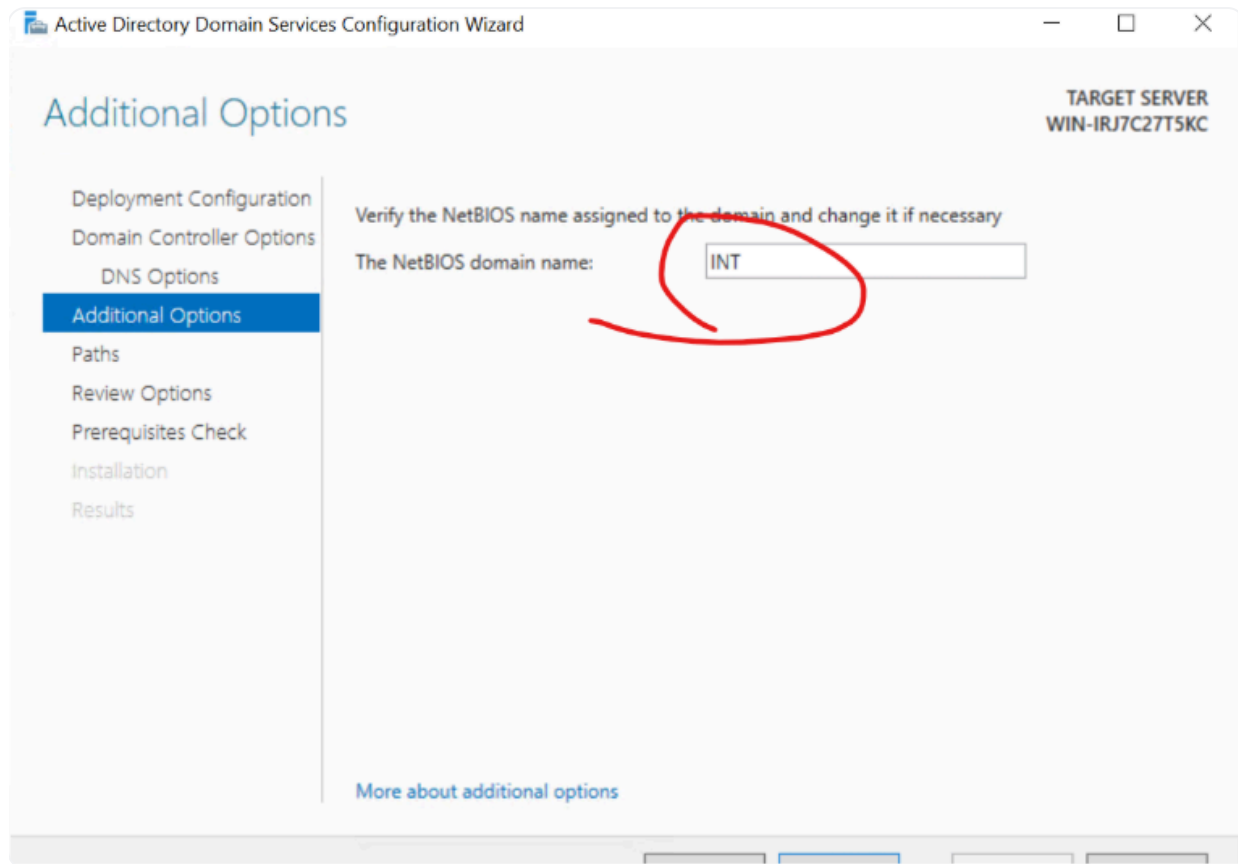
< Previous

Next >

Install

Cancel

NETBIOS will provide the Domain name you first gave to the server that you own from your Domain Provider (EX: [int.com](#))



- **Make sure it goes to the correct name space**
- Lastly Provide the location path for your Database, Log files (Errors etc.) and the System volume

(**Example:** Change Log Files Folder to (**C:\Local Disk\PerfLogs**)

Active Directory Domain Services Configuration Wizard

Paths

TARGET SERVER

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder: C:\Windows\NTDS

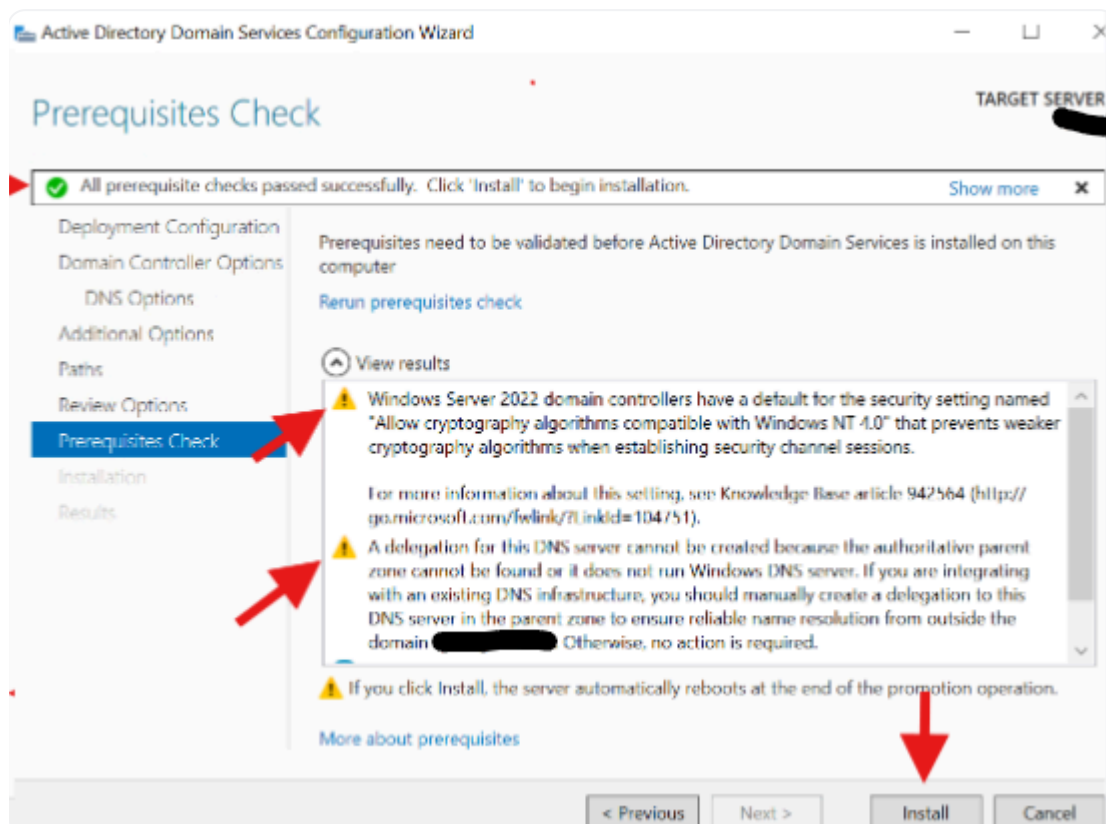
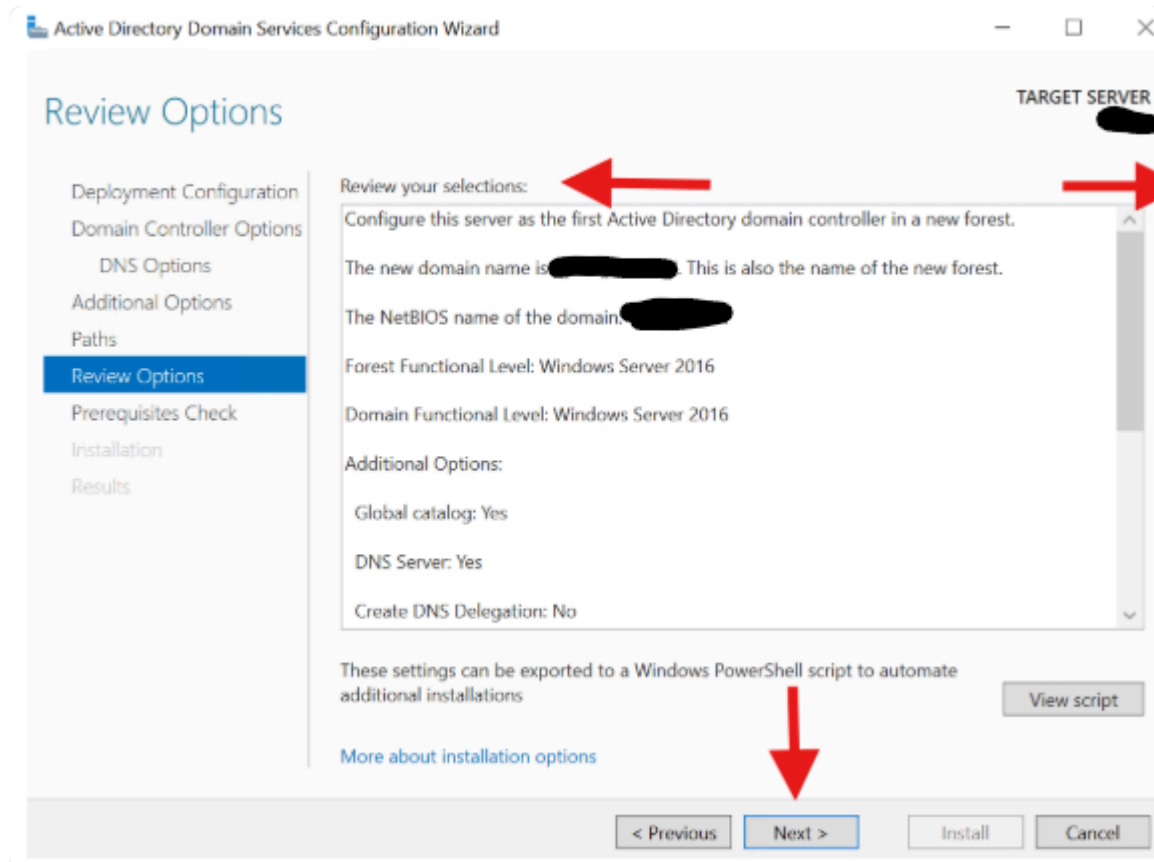
Log files folder: C:\Windows\NTDS

SYSVOL folder: C:\Windows\SYSVOL

[More about Active Directory paths](#)

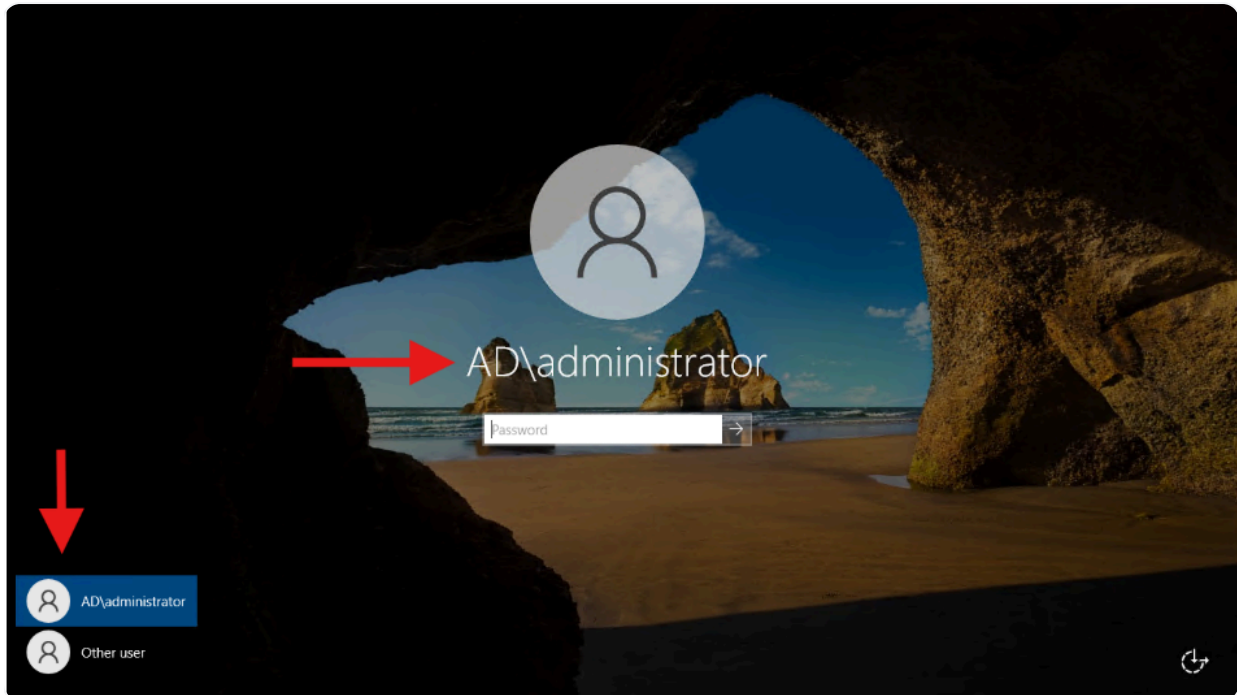
< Previous Next > Install Cancel

- Review your Selections for the wizard Then Begin the Prerequisite Checks for any errors to troubleshoot before continuing.
- If your Pre-requirements are met you will receive a checkmark to Install

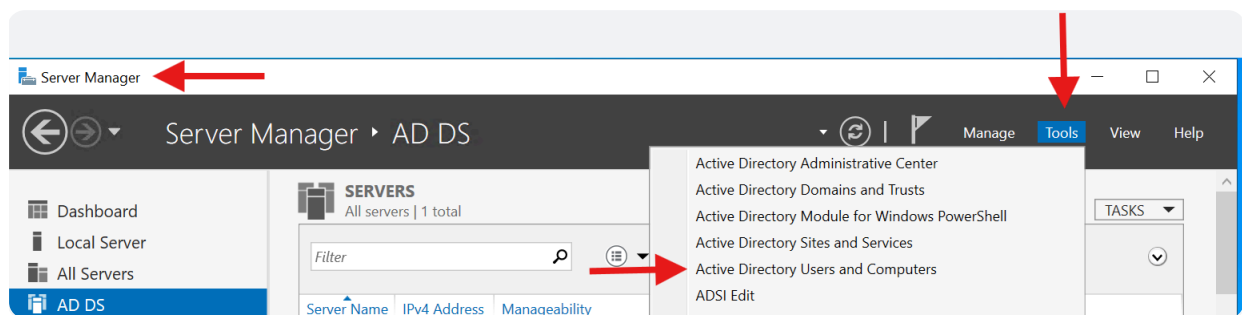


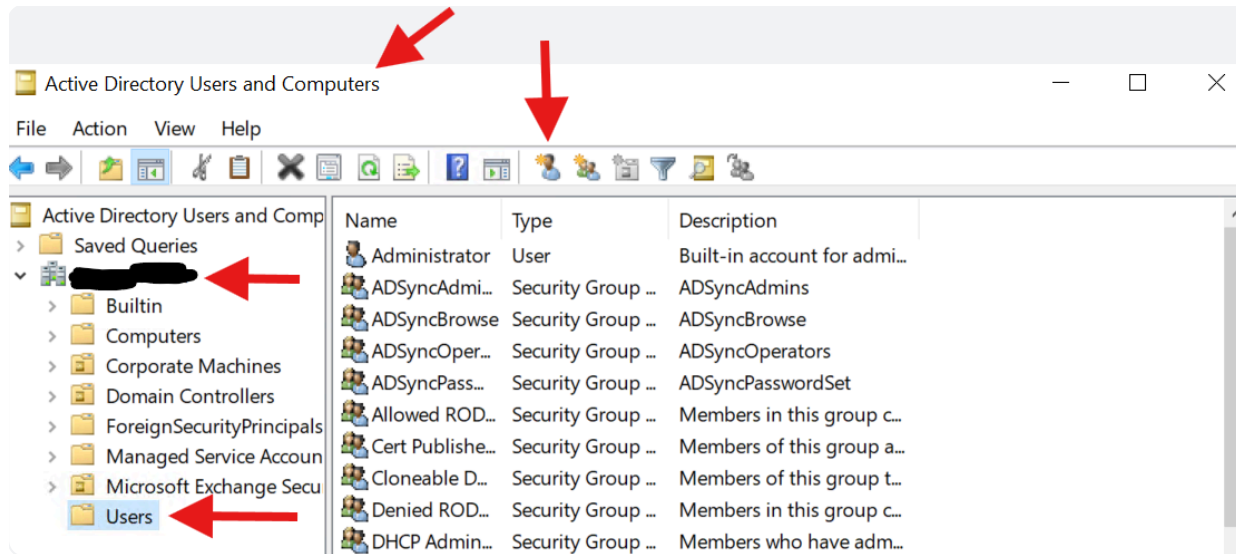
- After your Installation in Complete the Server will Sign out and will restart to add your New Server Roles.

- You will see the “AD” has been added as an Account. Before we are able to use Our new AD we will sign back in with **“Other user”** with the local machines credentials to **Create a New Admin user.**



- From **“Server Manager”** Go to **“Tools”** Then Select **“Active Directory Users and Computers”**
- From the new window for **“Active Directory Users and Computers”** go to your Domain Drop-down Then select Users From there Click the **“New User”** Icon indicated above





- Give your New **Admin/Users** name and the correct Domain they will logon to before Finishing the New user Object process. Provide the User Object a password Then confirm Any Additional Options like:
- **Having the user change the password on the next login / User cant change password / User Password Does not expire / or Disable the account Entirely for Security and Off-Boarding.**
- Under the new users Properties go to the “**Member Of**” Tab and Add Group and Security Service Principle Roles to the User Object (For this Example: **Enterprise Admin, Schema Admin, Domain Admin**) **ALWAYS PROVIDE USERS WITH LEAST PRIVILEGE TO THEIR ASSIGNED ROLES ON THE ORGANIZATION**
- Once added Click Apply You now have your new Admin/User with the Roles Needed for our **Microsoft Exchange Administrator**

New Object - User✕

 Create in: [redacted]/Users

First name:

David

Initials:

Last name:

Full name:

David

User login name:

David

@ [redacted] ▼

User login name (pre-Windows 2000):

[redacted]

David

< Back

Next >

Cancel

New Object - User



Create in: [REDACTED]/Users

Password:



.....

Confirm



.....

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires



☐ Account is disabled



< Back

Next >

Cancel

David Properties ? X

Remote control		Remote Desktop Services Profile			COM+
General	Address	Account	Profile	Telephones	Organization
Member Of		Dial-in	Environment		Sessions

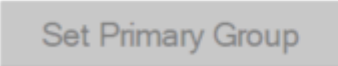
Member of:

Name	Active Directory Domain Services Folder
Domain Admins	[REDACTED]Users
Domain Users	[REDACTED]Users
Enterprise Admins	[REDACTED]Users
Schema Admins	[REDACTED]Users



Add... Remove

Primary group: Domain Users

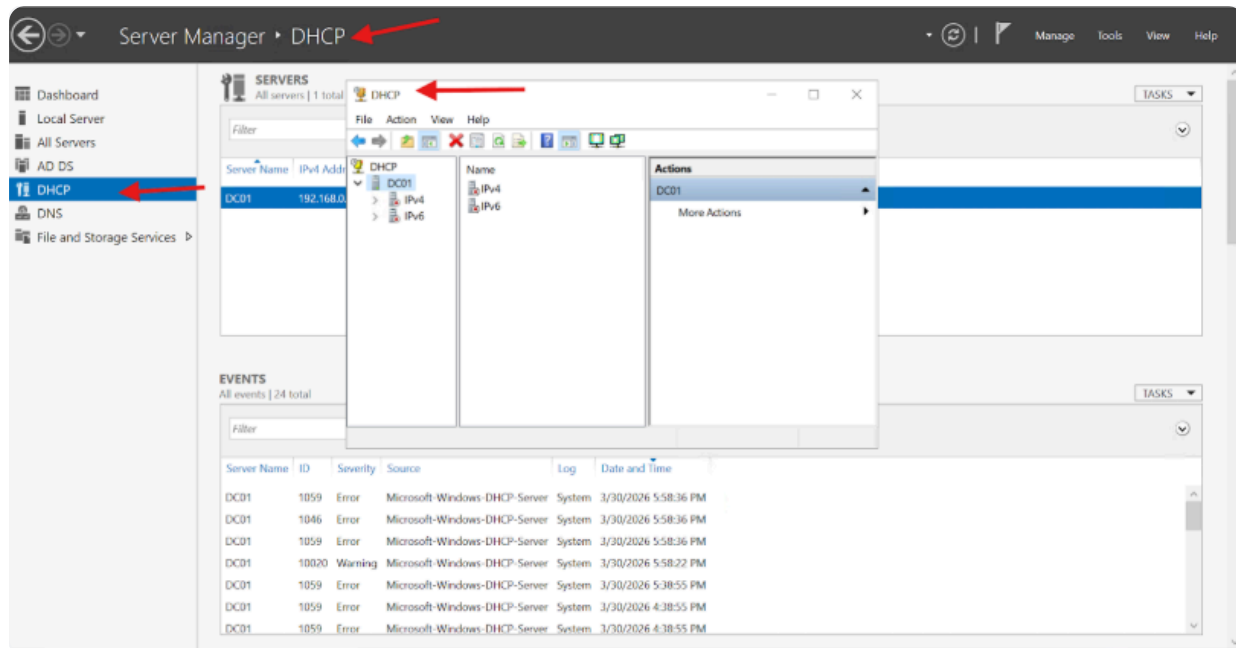
 There is no need to change Primary group unless you have Macintosh clients or POSIX-compliant applications.



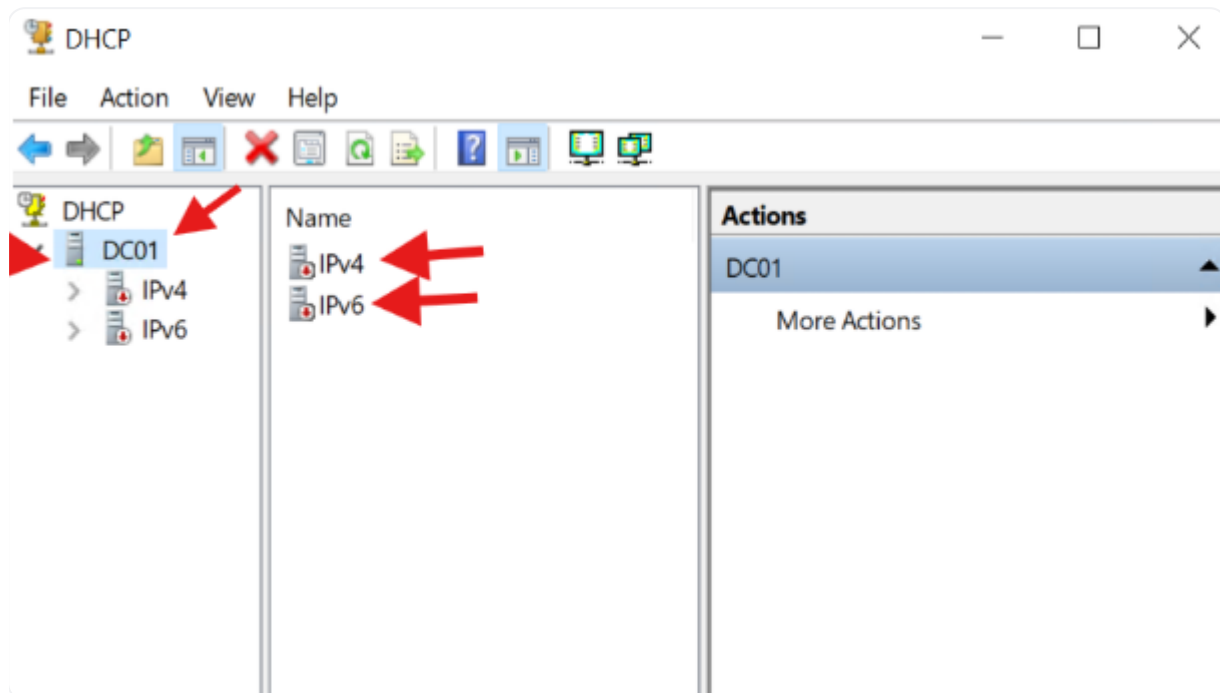
OK Cancel Apply Help

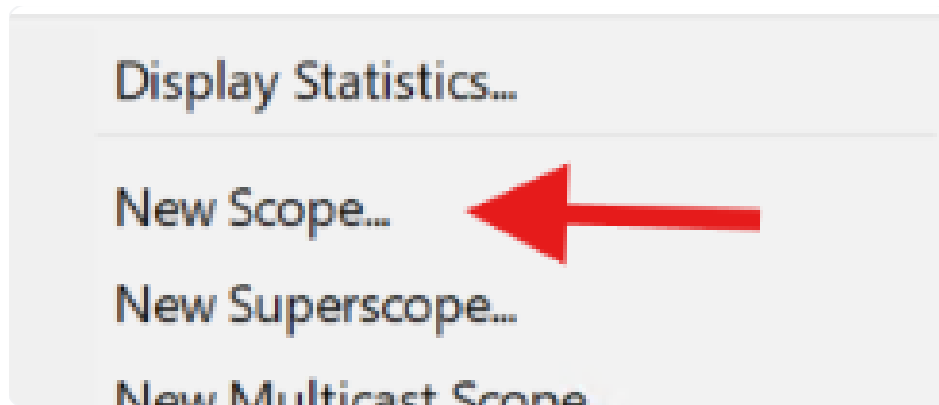
Section 2 Configure a DHCP Server

- Once Windows Server 2025/2022/2019/2016 Is Installed and Licensed Go to Add Server Roles and add the DHCP server role



- After DHCP has been added, Open DHCP from Server Manager to the New DHCP window. Under your DHCP you will see your SERVER (DC01) Then you will see a Drop-down for IPV4





- Clicking on IPV4 go to the Action Pane on the far right and select the drop down button New Scope give the Scope a Name (Ex:main Lan) and a Range number of network devices can be assigned an IP address (.3 - .254) 249 available address for computers. Set a Lease Duration for these IP addresses so all domain-joined computers can reach out to the DHCP server for a new IP address when expired.

A screenshot of the 'New Scope Wizard' dialog box. The title bar says 'New Scope Wizard'. The main heading is 'Scope Name'. Below it, a message says: 'You have to provide an identifying scope name. You also have the option of providing a description.' To the right of this message is a folder icon. Below the message, there is a text box for 'Name' containing 'Main.Lan' and a text box for 'Description' containing 'IP Ranges to Give out for all my AD Users and computers'. A red arrow points to the 'Name' text box. At the bottom of the dialog, there are three buttons: '< Back', 'Next >', and 'Cancel'. A red arrow points to the 'Next >' button.

Scope

Scope name: Main Lan

Start IP address: 192.168.1.3

End IP address: 192.168.1.254

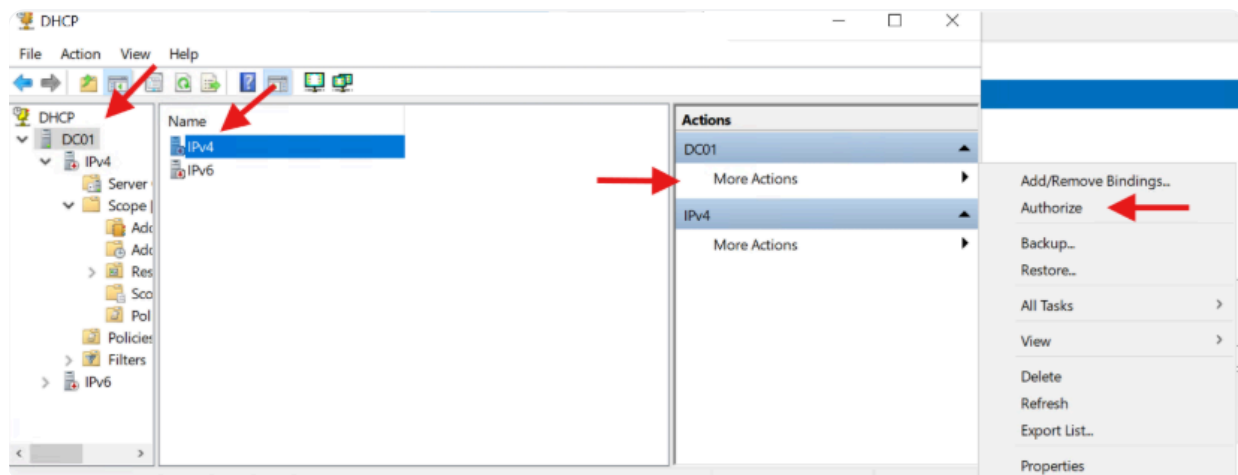
Subnet mask: 255.255.255.0 Length: 24

Lease duration for DHCP clients

☒ Limited to:

Days: 999 Hours: 23 Minutes: 9

☐ Unlimited



- After all additional information or desired configuration is met, return to your Domain Controller Icon (DC01) - go to the “More Actions” Drop-down and Select Authorize to Enable your DHCP server

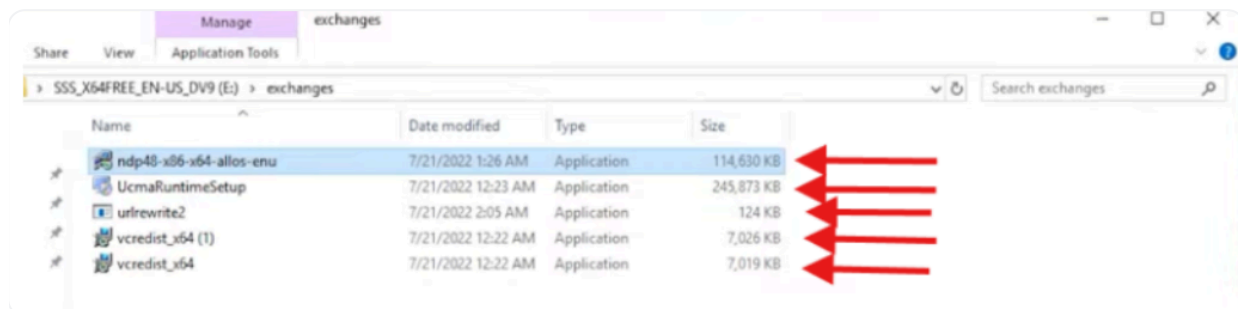
Section 3: Install an On-Premises Exchange Mail Server with "Mail Server" Role

- To Install Exchange Server 2019 OR SE Follow the Microsoft Requirements Page and System Requirements: Including Adding the Machine to your AD With your New Domain

- <https://learn.microsoft.com/en-us/exchange/plan-and-deploy/system-requirements?view=exchserver-2019>
- <https://learn.microsoft.com/en-us/exchange/plan-and-deploy/prerequisites?view=exchserver-2019>
- Install the Pre-requisite software Beginning with IIS Web Server to share information over the internet. Then begin installing all Necessary applications (vcredist 2012-2013, .NET framework, Unified Communications Managed API 4.0, and IIS URL Rewrite module)



<https://learn.microsoft.com/en-us/exchange/plan-and-deploy/prerequisites?view=exchserver-2019>



The following Windows features must be installed via PowerShell before mounting the Exchange ISO

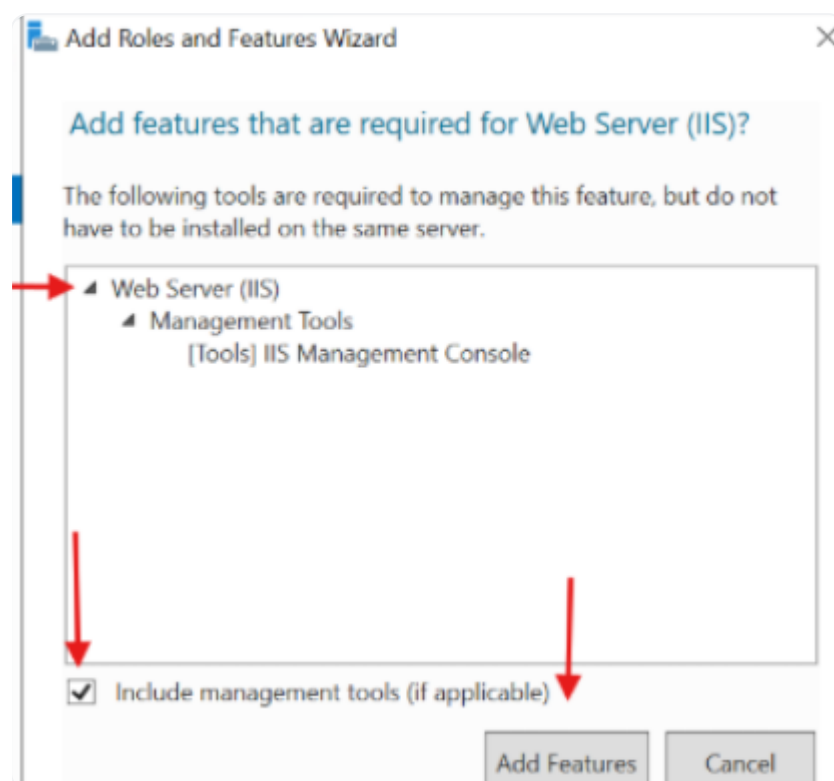
-Install the Remote Server Administration Tools (RSAT) for Active Directory Domain Services (ADDS) by running the following command in Windows PowerShell:

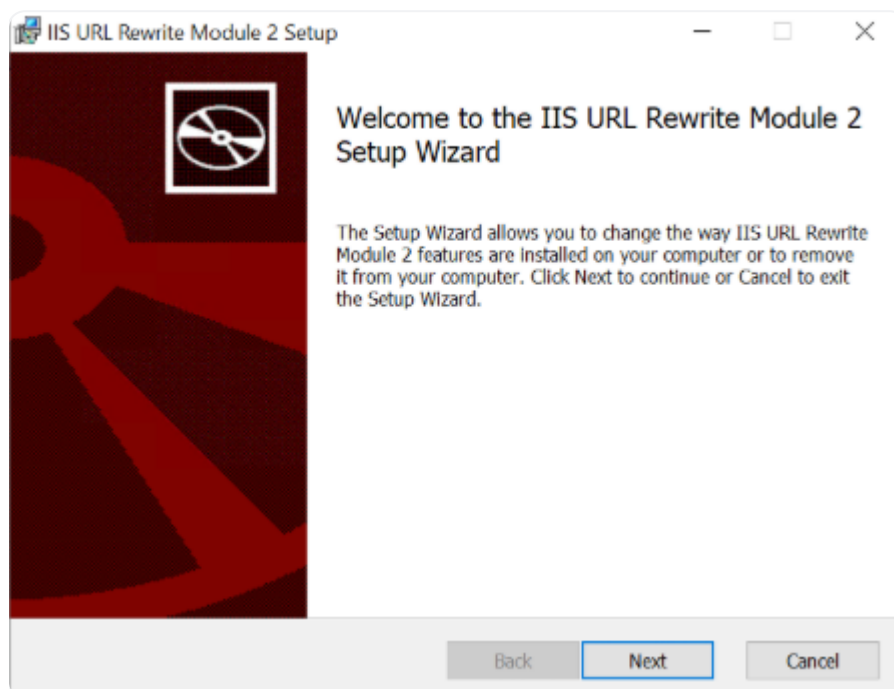
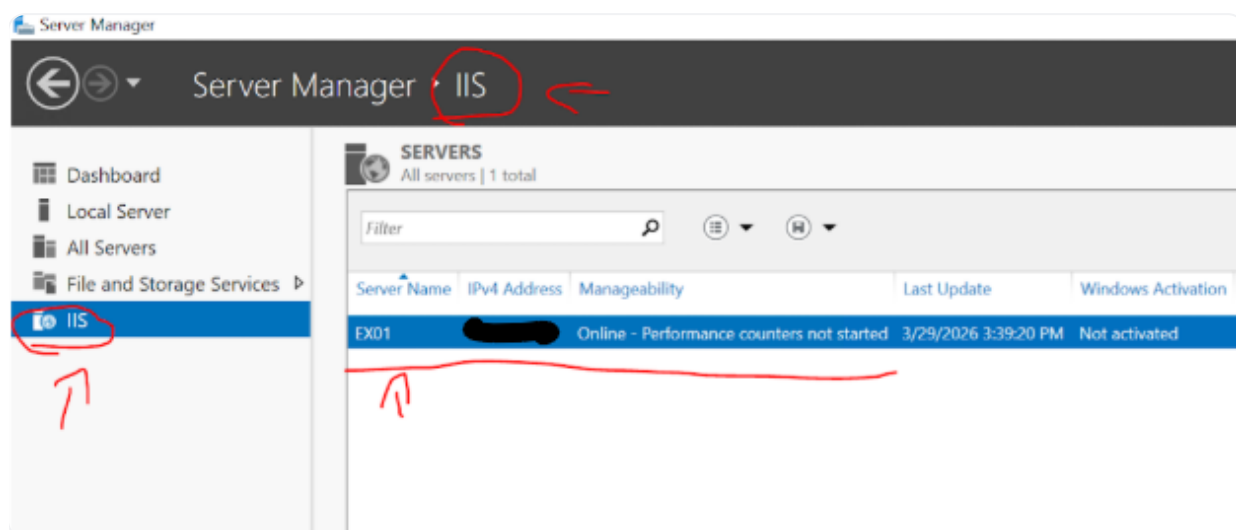
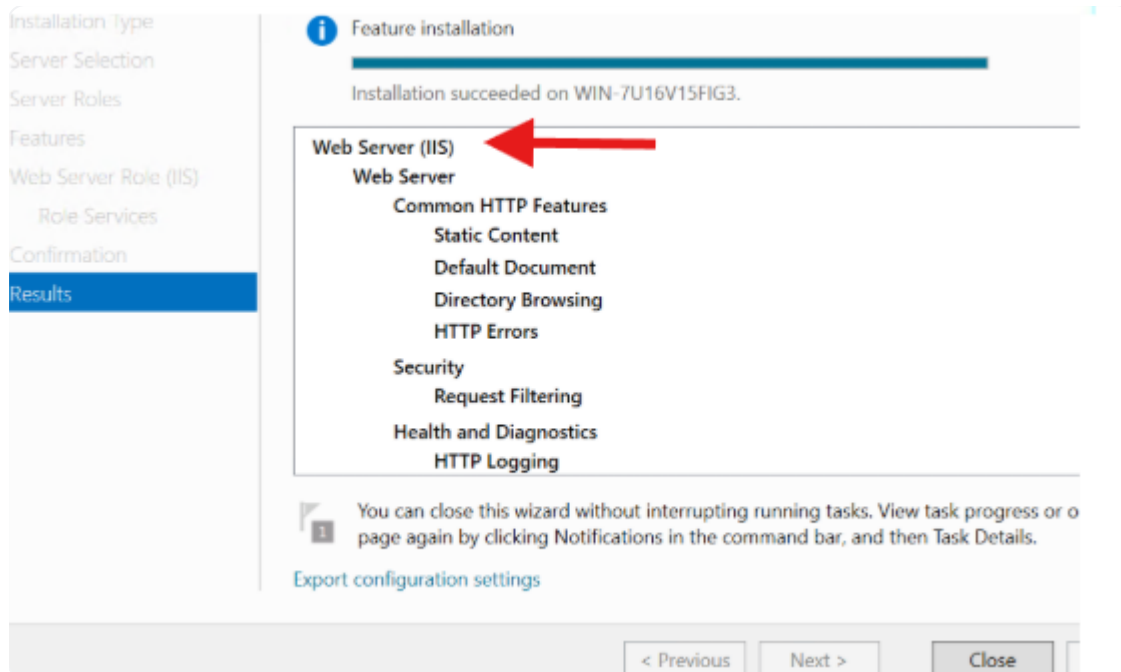
"Install-WindowsFeature RSAT-ADDS"

If you aren't going to use Exchange Setup to install the required Windows components (in the wizard or from the command line), run the one of the following commands in Windows PowerShell:

"Install-WindowsFeature Server-Media-Foundation, NET-Framework-45-Core, NET-Framework-45-ASPNET, NET-WCF-HTTP-Activation45, NET-WCF-Pipe-Activation45, NET-WCF-TCP-Activation45, NET-WCF-TCP-PortSharing45, RPC-over-HTTP-proxy, RSAT-Clustering, RSAT-Clustering-CmdInterface, RSAT-Clustering-Mgmt, RSAT-Clustering-PowerShell, WAS-Process-Model, Web-Asp-Net45, Web-Basic-Auth, Web-Client-Auth, Web-Digest-Auth, Web-Dir-Browsing, Web-Dyn-Compression, Web-Http-Errors, Web-Http-Logging, Web-Http-Redirect, Web-Http-Tracing, Web-ISAPI-Ext, Web-ISAPI-Filter, Web-Metabase, Web-Mgmt-Console, Web-Mgmt-Service, Web-Net-Ext45, Web-Request-Monitor, Web-Server, Web-Stat-Compression, Web-Static-Content, Web-Windows-Auth, Web-WMI, Windows-Identity-Foundation, RSAT-ADDS"

Lastly Make sure you Install IIS Services, Then perform an IIS ReWrite before finishing





Restart The Machine After Installing all the Pre-requisite Applications for Exchange

-Mount your Exchange 2019 or Exchange SE ISO



-Prepare All Aspects with an Exchange unattended install

<https://learn.microsoft.com/en-us/exchange/plan-and-deploy/prerequisites?view=exchserver-2019>

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> cd D:/
PS D:\> .\Setup.exe /IAcceptExchangeServerLicenseTerms_DiagnosticDataON /PrepareSchema
Microsoft Exchange Server 2019 Cumulative Update 15 Unattended Setup
Copying Files...
File copy complete. Setup will now collect additional information needed for installation.

Performing Microsoft Exchange Server Prerequisite Check
Prerequisite Analysis COMPLETED

Configuring Microsoft Exchange Server
Extending Active Directory schema COMPLETED

The Exchange Server setup operation completed successfully.

PS D:\> .\Setup.exe /IAcceptExchangeServerLicenseTerms_DiagnosticDataON /PrepareAD /OrganizationName:
Microsoft Exchange Server 2019 Cumulative Update 15 Unattended Setup
Copying Files...
File copy complete. Setup will now collect additional information needed for installation.

Performing Microsoft Exchange Server Prerequisite Check
Prerequisite Analysis 100%

Setup will prepare the organization for Exchange Server 2019 by using 'Setup /PrepareAD'. No Exchange Server 2016 roles have been detected in this topology. After this operation, you
will not be able to install any Exchange Server 2016 roles.
For more information, visit: https://learn.microsoft.com/Exchange/plan-and-deploy/deployment-ref/ms-exch-setupreadiness-NoE16ServerWarning?view=exchserver-2019

Setup will prepare the organization for Exchange Server 2019 by using 'Setup /PrepareAD'. No Exchange Server 2013 roles have been detected in this topology. After this operation, you
will not be able to install any Exchange Server 2013 roles.
For more information, visit: https://learn.microsoft.com/Exchange/plan-and-deploy/deployment-ref/ms-exch-setupreadiness-NoE15ServerWarning?view=exchserver-2019

Configuring Microsoft Exchange Server
Organization Preparation COMPLETED

Exchange Server setup operation completed successfully.
```

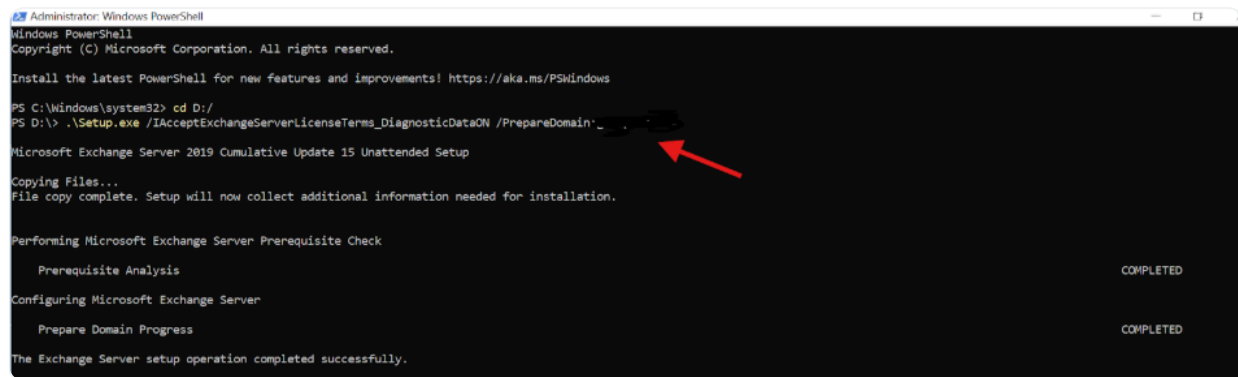
Find Which Drive Letter is used for the Mounted “Exchange 2019/Exchange SE”
And begin Preparing the AD Schema.

EXAMPLE: D:\Setup.exe

/IAcceptExchangeServerLicenseTerms_DiagnosticDataON /PrepareSchema

Once the Schema Had been Complete begin to Prepare the AD with this cmdlet

**E:\Setup.exe /IAcceptExchangeServerLicenseTerms_DiagnosticDataON
/PrepareAD /OrganizationName:"Contoso Corporation"**



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> cd D:/
PS D:\> .\Setup.exe /IAcceptExchangeServerLicenseTerms_DiagnosticDataON /PrepareDomain_...
Microsoft Exchange Server 2019 Cumulative Update 15 Unattended Setup

Copying Files...
File copy complete. Setup will now collect additional information needed for installation.

Performing Microsoft Exchange Server Prerequisite Check
    Prerequisite Analysis COMPLETED

Configuring Microsoft Exchange Server
    Prepare Domain Progress COMPLETED

The Exchange Server setup operation completed successfully.
```

Then Finally Prepare Your Domain with this Cmdlet

**E:\Setup.exe /IAcceptExchangeServerLicenseTerms_DiagnosticDataON
/PrepareDomain:**

**NOW YOU MAY BEGIN THE EXCHANGE 2019 INSTALL FROM YOUR MOUNT
DISK "SETUP.EXE"**

**Accept the License Agreement and choose rather you would like to share
diagnostic data with Microsoft.**

License Agreement

Please read and accept the Exchange Server license agreement.

MICROSOFT SOFTWARE LICENSE TERMS

MICROSOFT EXCHANGE SERVER 2019 STANDARD, ENTERPRISE, TRIAL, AND HYBRID

These license terms are an agreement between Microsoft Corporation (or based on where you live, one of its affiliates) and you. Please read them. They apply to the software named above, which includes the media on which you received it, if any. The terms also apply to any Microsoft updates, supplements, Internet-based services, and support services for this software unless other terms accompany those items. If so, those terms apply. Your use of this software may be subject to the additional agreements Microsoft has with you or your organization.

By using the software, you accept these terms. If you do not accept them, do not use the software. Instead, return it for a refund or credit. If you cannot obtain a refund there, contact Microsoft or the Microsoft affiliate serving your country for information about Microsoft's refund policies. See <https://www.microsoft.com/worldwide>.

As described below, using the software also operates as your consent to the transmission of certain computer information for Internet-based services.

If you comply with these license terms, you have the perpetual rights below.

- ☒ I accept the license agreement and will share diagnostic data with Microsoft (recommended).
- ☐ I accept the license agreement, but I'm not ready to share diagnostic data with Microsoft.
- ☐ I do not accept the license agreement.

[Learn more](#)

Exchange

next

Recommended Settings

- ☒ Use recommended settings

Exchange server will automatically check online for solutions when encountering errors and provide usage feedback to Microsoft to help improve future Exchange features.

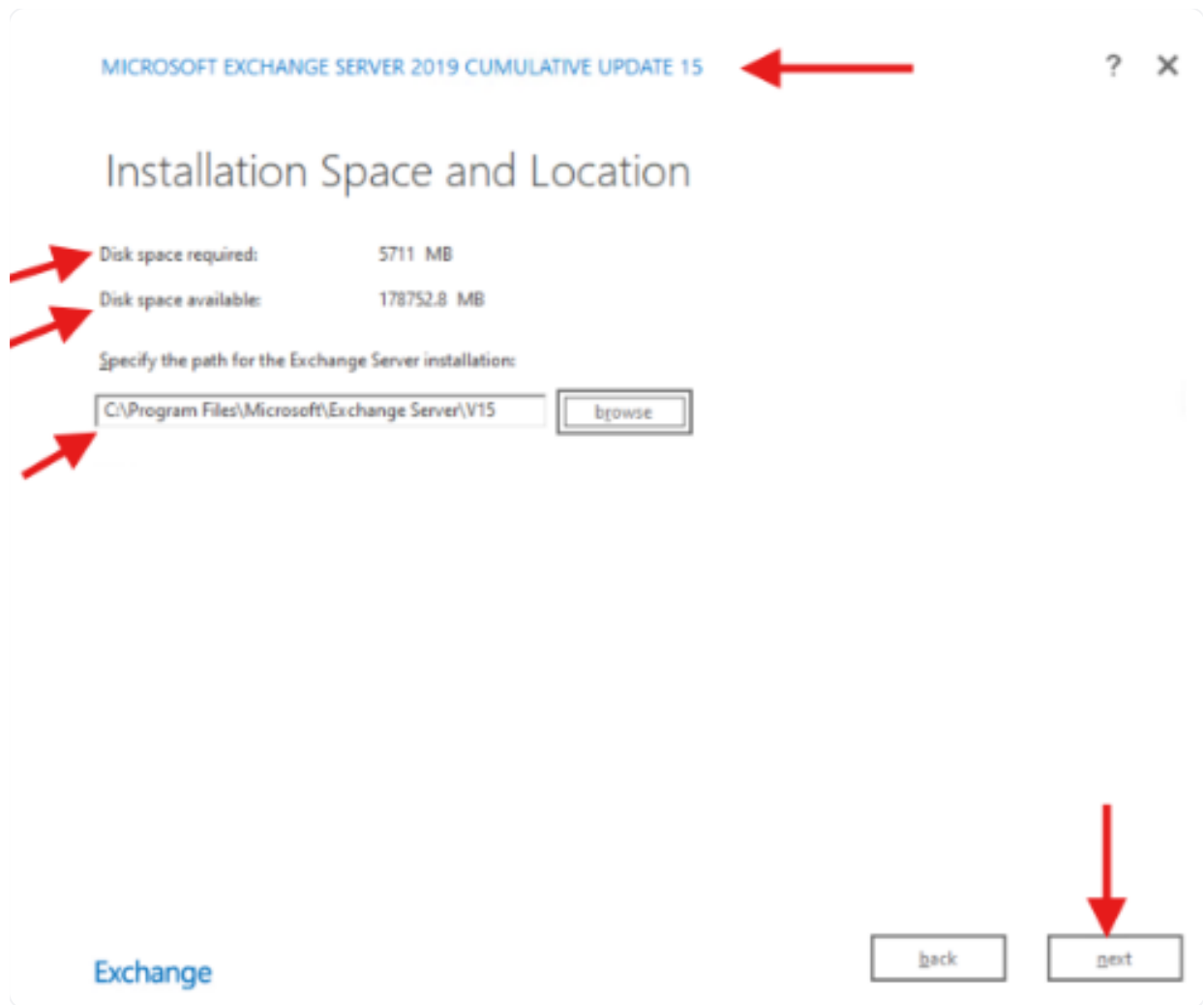
- ☐ Don't use recommended settings

Manually configure these settings after installation is complete (see help for more information).

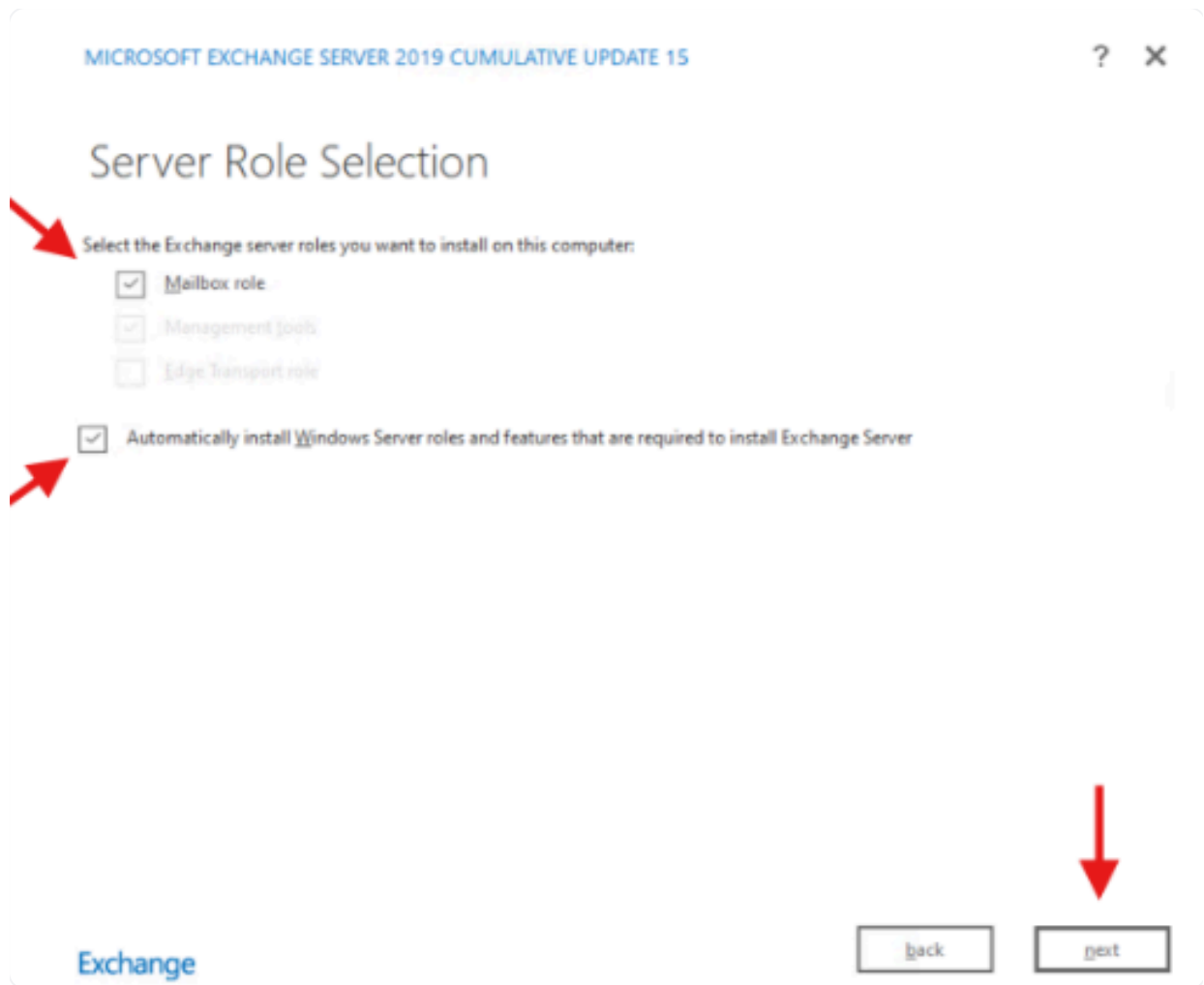
[Read more about providing usage feedback to Microsoft](#)

[Read more about checking for error solutions online](#)

MAKE SURE TO PLACE THE LOCATION OF EXCHANGE Install WITHIN THE C:DRIVE

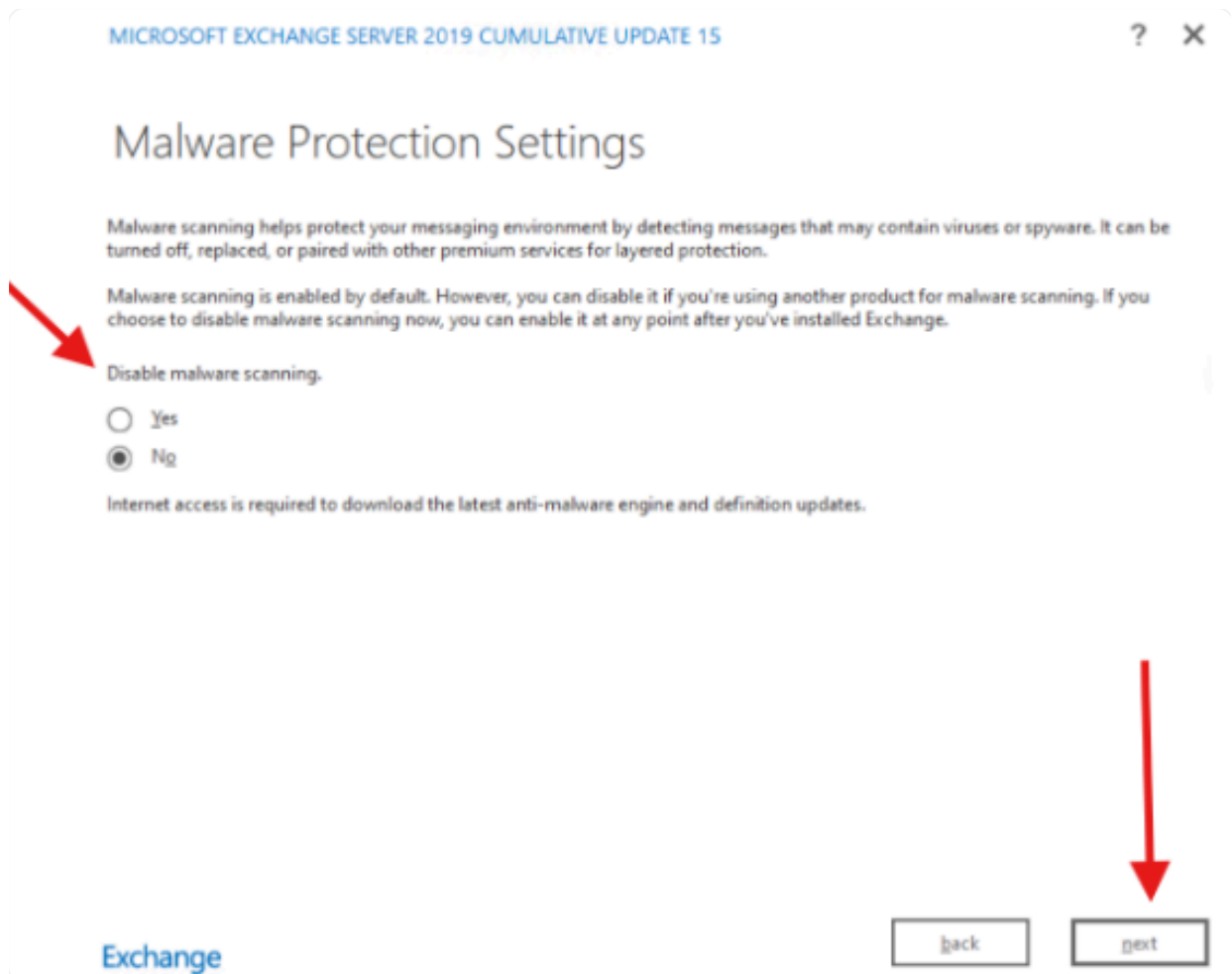


-Install your Exchange Server 2019 (Checkbox Which Roles you Need (In this case Mailbox/ Management Tools)



THIS IS OPTIONAL “Automatic install Windows Server roles and Features that are required to install Exchange Server”

Exchange Server 2019 includes built-in malware protection that scans all inbound, outbound, and internal emails for viruses and spyware. It's enabled by default on Mailbox servers and lets admins remove infected messages or attachments and set custom alerts.



-It is recommended you use third-party modern Email and Network security option for your Exchange and Exchange M365 Such as Proofpoint (Cloud), (kaspersky security, BitDefender Gravity Zone for Exchange or Using the Built-in Office 365 Defender(Cloud))

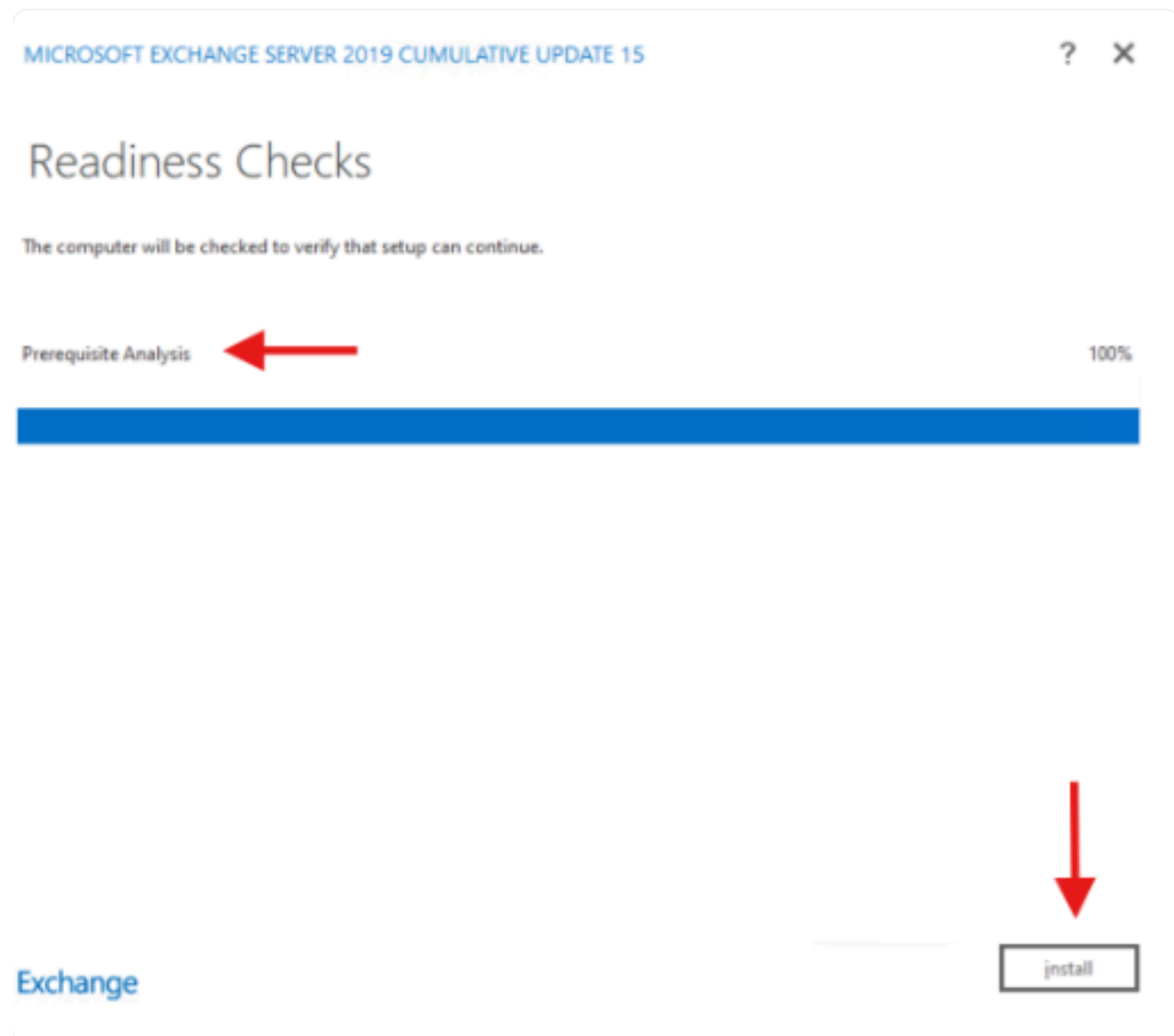


proofpoint.



Exchange
Online Protection

Watch for Any Setup errors these will show up in a Dumpfile with all errors during install and NOTE Exchange can get stuck around 16% if it has been over 2-3 hours on this progression try re-install the “Setup.exe”



Setup Progress



Step 2 of 13: Copy Exchange Files

16%



Exchange

Once Install Finishes Open an Internet Explorer (NOT EDGE) browser to access your Exchange Admin Center (Trusted Sites must be added to allow access to the site)

Setup Completed

Congratulations! Setup has finished successfully. To complete the installation of Microsoft Exchange Server, reboot the computer.

You can view additional post-installation tasks online by clicking the link: <https://aka.ms/ExchangePostInstallTasks>.

Exchange Setup has enabled Extended Protection on all required virtual directories on this machine. For more information, visit <https://aka.ms/EnableEPviaSetup>.

We recommend periodically running the Exchange Health Checker script, as well as after setup, to evaluate the organization for additional recommended configurations and security best practices. The Health Checker script can be downloaded from <https://aka.ms/ExchangeSetupHC>.

You can also start the Exchange Administration Center after Setup is finished.

☒ Launch Exchange Administration Center after finishing Exchange setup.

Exchange

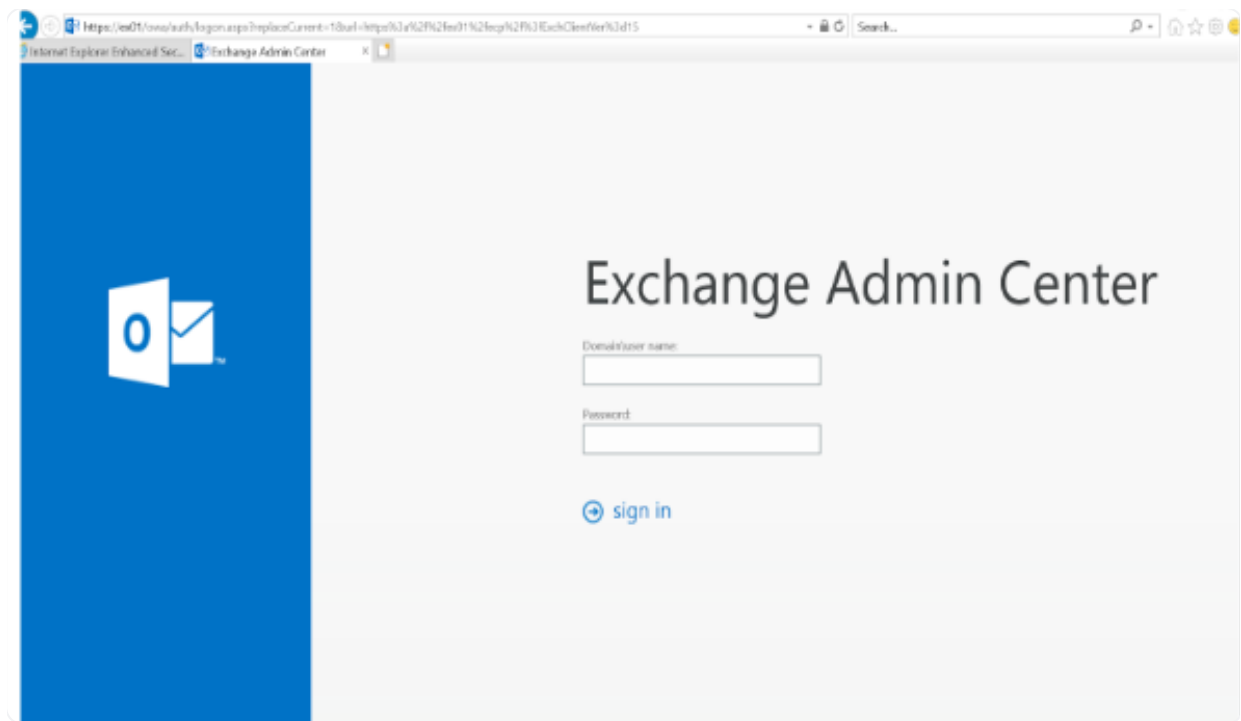
finish

Signin to your new Exchange Admin Center with your Exchange Admin Credentials

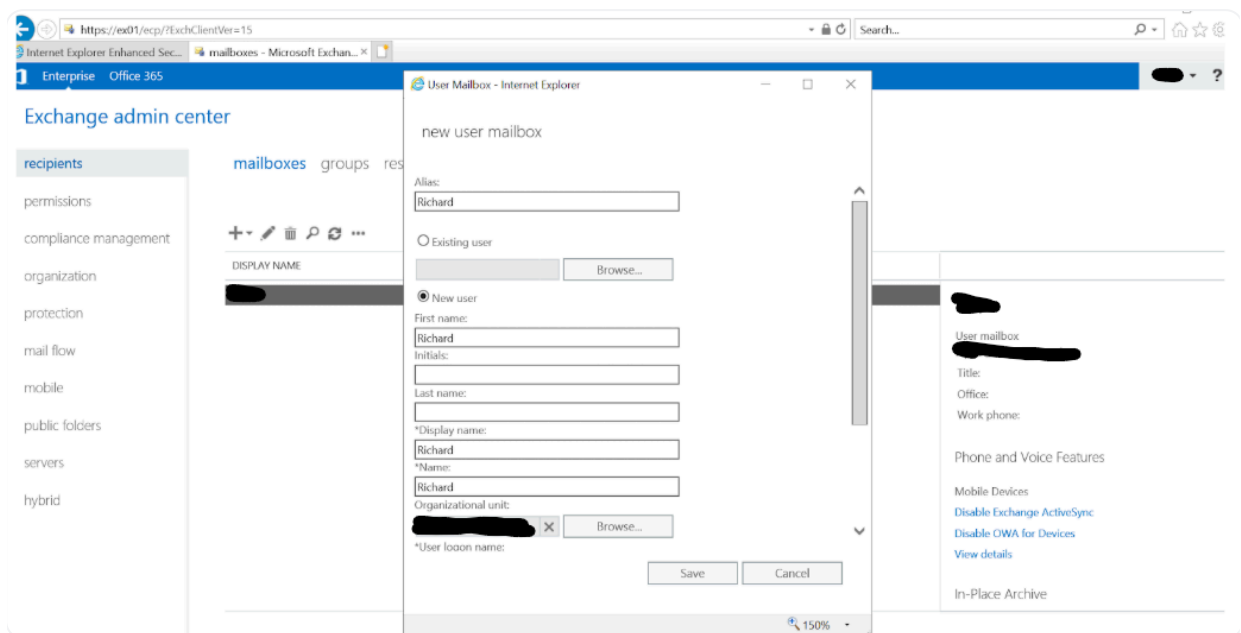
“Domain\User name”

“Password”

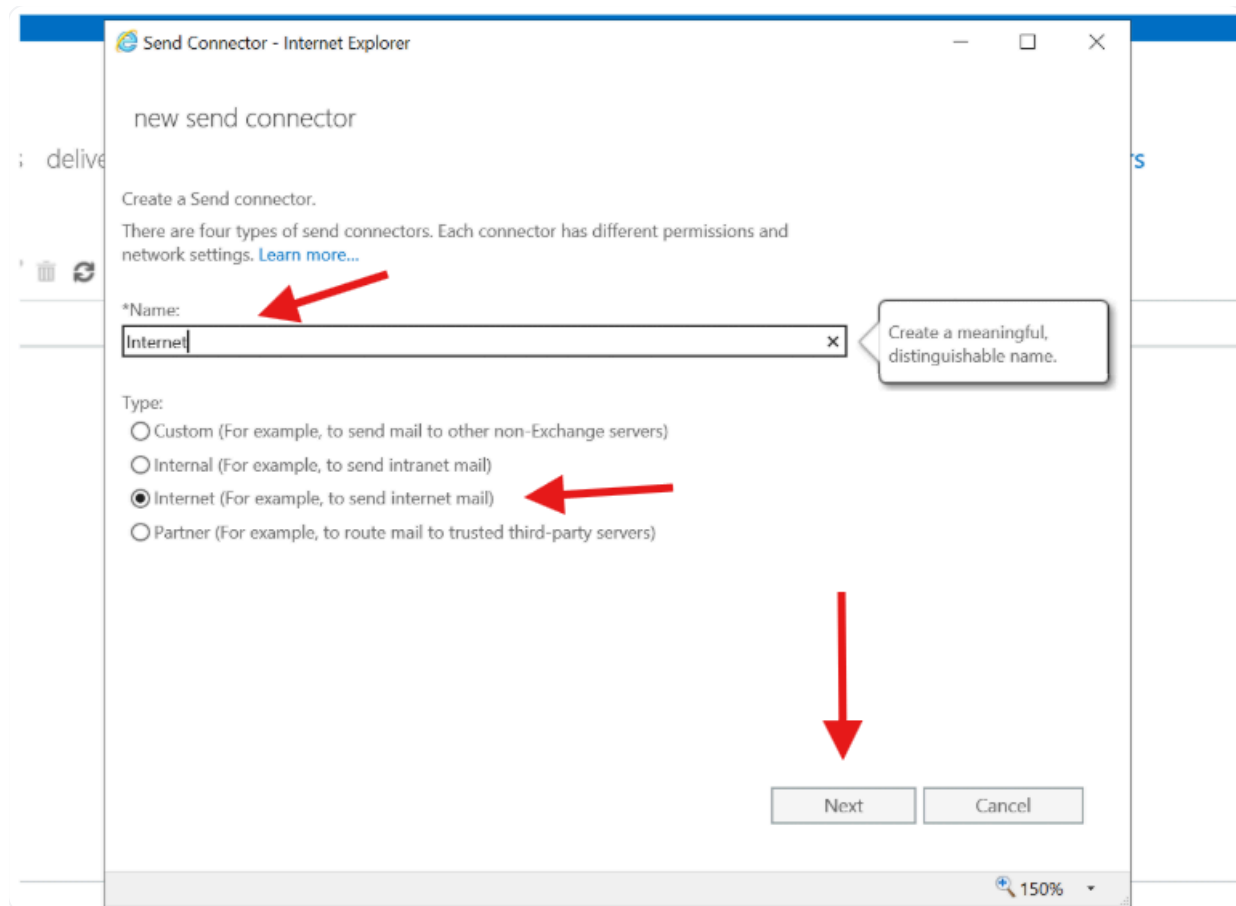
EX : [Contoso.com](#)\Richard



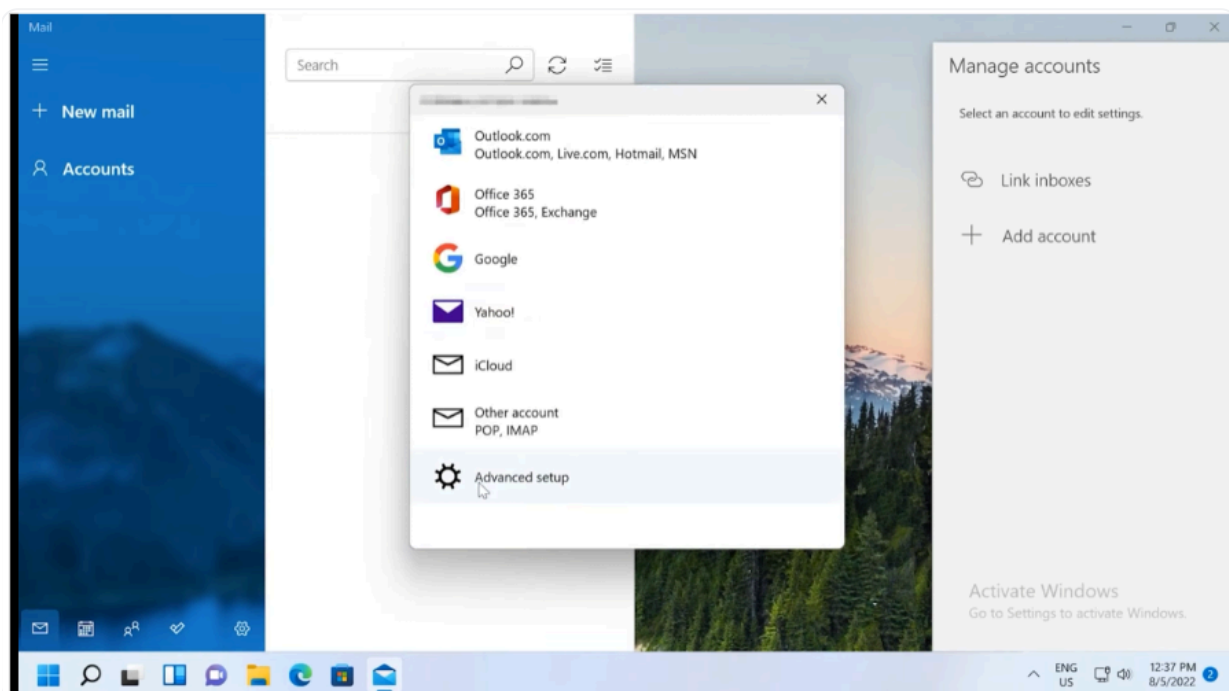
Create New User Mailbox for your Organization (Here ex: Richard)



Create a Send Connector for you to Send out email to the internet



Lastly Go to Sign-in to the newly created users account from a separate machine



Add the Account manually with Exchange ActiveSync

Advanced setup

Choose the kind of account you want to setup. If you're not sure, check with your service provider.

Exchange ActiveSync

Includes Exchange and other accounts that use Exchange ActiveSync.

Internet email

POP or IMAP accounts that let you view your email in a web browser.



✕ Cancel

Exchange

Email address

someone@example.com

Password

User name

Domain

Server



✓ Sign in ✕ Cancel

THEN LASTLY SIGN-IN TO THE USERS OUTLOOK ACCOUNT

